

jina-embeddings-v4: Universal Embeddings for Multimodal Multilingual Retrieval

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Abstract

We introduce [jina-embeddings-v4](#), a 3.8 billion parameter multimodal embedding model that unifies text and image representations through a novel architecture supporting both single-vector and multi-vector embeddings in the late interaction style. The model incorporates task-specific Low-Rank Adaptation (LoRA) adapters to optimize performance across diverse retrieval scenarios, including query-document retrieval, semantic text similarity, and code search. Comprehensive evaluations demonstrate that [jina-embeddings-v4](#) achieves state-of-the-art performance on both single-modal and cross-modal retrieval tasks, with particular strength in processing visually rich content such as tables, charts, diagrams, and mixed-media formats. To facilitate evaluation of this capability, we also introduce Jina-VDR, a novel benchmark specifically designed for visually rich image retrieval.

1 Introduction

We present [jina-embeddings-v4](#), a multimodal embedding model capable of processing text and image data to produce semantic embedding vectors of varying lengths, optimized for a broad array of applications. It incorporates optimized LoRA adapters [Hu et al., 2022] for information retrieval and semantic text similarity. An adapter is also provided for programming language embeddings, technical question-answering, and natural language code retrieval. It also brings new functionality to processing visually rich images (also called visual documents), i.e., materials mixing texts and images, containing tables, charts, diagrams, and other kinds of common mixed media [Ding et al., 2024]. We have also developed Jina-VDR, a new multilingual, multi-domain benchmark suite for a broad range of visual retrieval tasks, to evaluate the capabilities of [jina-embeddings-v4](#).

We discuss the challenges of developing a multimodal, multi-functional, state-of-the-art embedding model capable of handling texts in a variety of languages, including computer coding languages, images, and “visually rich” data. The resulting model, [jina-embeddings-v4](#), projects inputs from all modalities into a unified semantic space, minimizing or eliminating the “modality gap” that has troubled similar projects [Liang et al., 2022a]. In addition, we introduce Jina-VDR, an advanced benchmark for images like screenshots and scans of visually complex documents.

The major contributions of this work are as follows:

- We introduce a unified multi-task learning paradigm that jointly optimizes embedding models to represent texts and images as single- and multi-vector embeddings.
- Building on work done for [jina-embeddings-v3](#), we train LoRA extensions to enhance support for specific domains and task types, achieving results comparable to specialized models.
- We have made particularly strong progress in handling visually rich images, especially for tasks outside of the existing ViDoRe benchmark [Faysse et al., 2025], which is limited to question-answering. [jina-embeddings-v4](#) outperforms other multimodal models by a significant margin on this type of material and supports a much more diverse set of use scenarios.
- We construct a multilingual, multi-domain benchmark for screenshot retrieval. In contrast to other retrieval benchmarks (i.e., [Faysse et al., 2025, Xiao et al., 2025]) that focus on question answering and OCR-related tasks, we expand the scope of visual document benchmarking to multilingual retrieval, more query types, and a much more diverse

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array of materials, like maps, diagrams, advertisements, and other mixed media.

2 Background

The underlying principles behind embedding models are indifferent to data modality. An embedding model transforms digitally encoded objects into vectors in a high-dimensional embedding space such that some of the semantic features of the objects, depending on the model’s training regimen, correspond to subspaces in that embedding space. Objects with more such features in common will have corresponding vectors that are closer to each other by some metric (typically cosine similarity) than objects with fewer common features.

Individual models, however, only support the modalities for which they are designed and trained. Embedding models initially developed primarily to support natural language texts, like *jina-embeddings-v3* [Sturua et al., 2024], but there are many image embedding models, and more recently, audio and video models. The semantic embedding paradigm can also encompass models that support more than one modality, like bimodal image-text models, including OpenAI’s CLIP [Radford et al., 2021] and subsequent developments including *jina-clip* [Koukounas et al., 2024]. The principal purpose of multimodal embedding models is to project objects from multiple modalities into the same semantic embedding space, so that, for example, a picture of a cat and a text discussing or describing a cat will correspond to relatively close embedding vectors.

Embedding models can also specialize in specific types of input within a single modality. There are text embedding models designed for programming code [Liu et al., 2024], legal texts [Voyage AI, 2024], and other special domains. There is also recent work in specialized image embedding models designed to support “visually rich” data, such as screenshots, charts, and printed pages that combine text and imagery and have internal visual structure [Faysse et al., 2025, Ma et al., 2024].

There are other dimensions of embedding model specialization as well. Models can be optimized for specific tasks, such as information retrieval, clustering, and classification [Sturua et al., 2024]. They can also vary based on the nature of the embeddings they produce. Most are single-/dense vector models, generating one embedding vector for whatever input they are given. There are also multi-vector/late

interaction models, such as ColBERT [Khattab and Zaharia, 2020] and ColPali [Faysse et al., 2025]. Late interaction is generally a more precise measure of semantic similarity for retrieval, but has significantly greater storage and computing costs.

Instead of specializing, *jina-embeddings-v4* builds on a single base model to provide competitive performance as a text, image, and cross-modal embedding model with strong performance handling visually rich documents. It supports both single-vector and multi-vector and is optimized to provide embeddings of varying lengths. Furthermore, the model includes LoRA extensions that optimize it for specific application classes: information retrieval, multimodal semantic similarity, and computer code retrieval.

This single-model approach entails significant savings in practical use cases when compared to deploying multiple AI models for different tasks and modalities.

3 Related Work

Transformer-based neural network architectures that generate semantic embeddings are well-established [Reimers and Gurevych, 2019], and there is a sizable literature on training techniques for them. Multi-stage contrastive training [Wang et al., 2022], and techniques for supporting longer texts [Günther et al., 2023] are particularly relevant to this work.

Compact embedding vectors bring valuable performance benefits to AI applications, and this motivates work in Matryoshka Representational Learning (MRL) [Kusupati et al., 2022] as a way to train models for truncatable embedding vectors.

Contrastive text-image training has led to groundbreaking results in zero-shot image classification and cross-modal retrieval in conjunction with dual encoder architectures like CLIP [Radford et al., 2021]. However, recent work shows better performance from vision-language models (VLMs) like Qwen2.5-VL-3B-Instruct [Bai et al., 2025]. Jiang et al. [2024] show that VLMs suffer less from a modality gap than dual encoder architectures.

In contrast to [Zhang et al., 2024], *jina-embeddings-v4* is trained on multilingual data, supports single as well as multi-vector retrieval, and does not require task-specific instructions. Other VLM models are trained exclusively on data for text-to-image [Faysse et al., 2025, Ma et al., 2024] or text-to-text retrieval [Jiang et al., 2024].